

# Sharpness of Irrationality after One Stabilization for Cubic Threefolds

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## Abstract

The irrationality of  $X \times \mathbf{P}^1$  for every smooth complex cubic threefold  $X$  is sharp: two explicit smooth cubic threefolds over  $\mathbf{Q}$  are rational after multiplication by  $\mathbf{P}^2$ . For each, the least  $m$  such that  $X \times \mathbf{P}^m$  is rational is two, over both  $\mathbf{Q}$  and  $\mathbf{C}$ . We prove more generally that if  $k$  has characteristic zero,  $S$  is a smooth quartic del Pezzo surface with  $S(k) \neq \emptyset$ , and  $\text{Pic}(S_{\bar{k}})$  is a stably permutation Galois module, then  $S \times \mathbf{A}^2$  is  $k$ -rational. We identify a rational quotient of the projective Cox model by a saturated rank-three subtorus; the residual two-dimensional torus is rational. Applying this theorem to the quartic del Pezzo fibrations of Tschinkel and Zhang, we prove that every member of both cubic series is rational after multiplication by  $\mathbf{P}^2$ , giving a uniform two-dimensional bound for their stable-rationality results. Consequently the stabilization level is two on explicit examples in cubic-threefold moduli and infinite at a very general point. We also obtain smooth projective nonrational fourfolds  $Y/\mathbf{Q}$  such that  $Y \times \mathbf{A}^1$  is rational.

**Keywords.** cubic threefold; quartic del Pezzo surface; stable rationality; universal torus; variety with one apparent double point; algebraic torus; rationality after affine-line stabilization.

## 1 Introduction and main results

For a projective variety  $V/k$ , put

$$\ell_k(V) = \min\{m \geq 0 : V \times \mathbf{A}^m \text{ is } k\text{-rational}\}.$$

If the set is empty, put  $\ell_k(V) = \infty$ . Since  $\mathbf{A}^m$  and  $\mathbf{P}^m$  are birational, either factor gives the same invariant. This is the level of stable rationality considered by Tschinkel and Zhang [4, p. 13].

By the one-stabilization theorem [3, Theorem 1.1], every smooth complex cubic threefold  $X$  satisfies  $\ell_{\mathbf{C}}(X) \geq 2$ . Tschinkel and Zhang constructed explicit smooth cubic threefolds over  $\mathbf{Q}$  that are stably rational [4, Theorem 1.3]. Their levels are finite, and rationality after base change gives  $\ell_{\mathbf{Q}}(X) \geq \ell_{\mathbf{C}}(X_{\mathbf{C}}) \geq 2$ . We prove equality for two of these threefolds.

Consider the smooth cubic threefolds

$$X_1 : (x_4 - 2x_3)x_1^2 + 3(x_4 + 2x_3)x_2^2 + 3x_3^2x_4 - x_4^3 + x_5^3 = 0, \quad (1)$$

$$X_3 : x_4(x_1^2 + 2x_1x_2) + x_3(x_1^2 + x_1x_2 + x_2^2) + x_3^3 - x_3x_4^2 + x_4^3 + 2x_5^3 = 0. \quad (2)$$

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**Theorem 1.1** (Exact level for the cubic threefolds). *For  $j \in \{1, 3\}$ ,*

$$\ell_{\mathbf{Q}}(X_j) = \ell_{\mathbf{C}}((X_j)_{\mathbf{C}}) = 2.$$

*Equivalently,  $(X_j)_{\mathbf{C}} \times \mathbf{P}^1$  is irrational, whereas  $X_j \times \mathbf{P}^2$  is rational over  $\mathbf{Q}$ .*

The following surface theorem supplies the upper bound in Theorem 1.1.

**Theorem 1.2** (Two-dimensional stabilization). *Let  $k$  be a field of characteristic zero and let  $S/k$  be a smooth quartic del Pezzo surface. Suppose that  $S(k) \neq \emptyset$  and that  $\text{Pic}(S_{\bar{k}})$  is a stably permutation  $\text{Gal}(\bar{k}/k)$ -module. Then*

$$S \times \mathbf{A}^2 \quad \text{and, equivalently,} \quad S \times \mathbf{P}^2$$

*are  $k$ -rational. In particular, this holds for every stably  $k$ -rational smooth quartic del Pezzo surface.*

From [4] we use the Cox model, tangent-projection theorem, Galois classification, and generic splitting of the universal torsor. We supply a rank-three quotient of the projective Cox model.

When  $S(k) \neq \emptyset$ , the hypothesis on  $\text{Pic}(S_{\bar{k}})$  is condition (1) of [4, Corollary 4.3]. Manin proves that stable rationality implies this condition [2, Theorem 2.2]; [4, Proposition 4.1 and Lemma 4.2] identify the remaining equivalent conditions. We prove that condition (1) implies the stronger conclusion that  $S \times \mathbf{A}^2$  is rational. The stable-permutation resolution in [4, p. 13] gives a uniform bound of eleven variables; the quotient construction gives two.

For  $r \geq 0$ , let  $X_{j,r} \subset \mathbf{P}_{\mathbf{Q}}^{r+4}$  be obtained from equation (1) or (2) by adding  $w_1^3 + \cdots + w_r^3$  to its left-hand side. Thus  $X_{j,0} = X_j$ .

**Corollary 1.3** (Two series of cubic hypersurfaces). *For every  $r \geq 0$  and  $j \in \{1, 3\}$ , the variety  $X_{j,r} \times \mathbf{P}^2$  is rational over  $\mathbf{Q}$ .*

Tschinkel and Zhang supply the smoothness, fibrations, and generic-fibre Galois types in [4, Propositions 5.1–5.2]. Applying Theorem 1.2 over the two base function fields proves Corollary 1.3 and sharpens their stable-rationality conclusions to  $X_{j,r} \times \mathbf{P}^2$  rational.

**Corollary 1.4** (Rationality after affine-line stabilization). *For  $j \in \{1, 3\}$ , the variety  $Y_j = X_j \times \mathbf{P}^1$  is a smooth projective fourfold over  $\mathbf{Q}$  which remains nonrational over  $\mathbf{C}$ , while  $Y_j \times \mathbf{A}^1$  is rational over  $\mathbf{Q}$ .*

The fourfolds  $Y_j$  answer the question posed in [4, p. 13]: find a nonrational variety  $Y$  such that  $Y \times \mathbf{A}^1$  is rational. This is a birational statement, not Zariski cancellation for isomorphic affine cylinders.

**Corollary 1.5** (Variation in the moduli of cubic threefolds). *On the moduli space of smooth complex cubic threefolds,  $\ell_{\mathbf{C}}$  is equal to two on every isomorphism class represented by  $X_1$  or  $X_3$ , and is infinite at a very general point.*

Here is the proof of the surface theorem. Let  $Z$  be the projective Cox model of  $S$ . The rank-five torus  $T_0$  acting on  $Z$  contains a saturated rank-three subtorus  $T_3$ . Four of its weight blocks form a Galois-stable set, and their weight differences form a lattice basis. A codimension-three tangent section meets a general  $T_3$ -orbit once, so tangent projection identifies  $Z/T_3$  birationally with  $\mathbf{P}^4$ . Generic splitting of the universal torsor also gives  $Z/T_3 \sim_k S \times (T_0/T_3)$ . The residual torus  $T_0/T_3$  has dimension two and is rational; the two descriptions of the quotient therefore give the rationality of  $S \times \mathbf{A}^2$ . Applying the argument over the two base function fields proves Corollary 1.3. In Section 2 we prove the quotient criterion, in Section 3 we construct the required Cox-model section, and in Section 4 we deduce the rationality statements.

## 2 A criterion for rational torus quotients

Let  $Z \subset \mathbf{P}(V)$  be a geometrically integral  $n$ -fold over an infinite field  $k$ . At a smooth point  $p \in Z(k)$ , write

$$\pi_p : Z \dashrightarrow \mathbf{P}(V/\widehat{T}_p Z) \quad (3)$$

for projection from the embedded tangent space; the hat denotes the affine cone. In the application,  $Z \subset \mathbf{P}^{2n+1}$  has one apparent double point, and  $\pi_p$  is birational for general  $p$  by [4, Theorem 2.4].

**Theorem 2.1** (A quotient criterion). *Let a rank- $r$   $k$ -torus  $T$  act projectively linearly and generically freely on  $Z \subset \mathbf{P}(V)$ , and suppose that the generic tangent projection  $\pi_p$  is birational to  $\mathbf{P}^n$ . Choose a diagonal linear lift of the action over  $k^s$ . Its weights are determined up to a common translation. Suppose that it has nonzero weight spaces  $V_{w_0}, \dots, V_{w_r}$  such that  $w_1 - w_0, \dots, w_r - w_0$  is a  $\mathbf{Z}$ -basis of  $X^*(T_{k^s})$ . Assume that the sum  $W = V_{w_0} \oplus \dots \oplus V_{w_r}$  and the sum  $\widehat{B}$  of all other weight spaces descend to complementary  $k$ -subspaces of  $V$ ; write  $B = \mathbf{P}(\widehat{B})$ .*

*Suppose that there is a point  $p \in Z(k)$  for which  $\pi_p$  is birational, together with a  $k$ -defined codimension- $r$  linear subspace*

$$\Lambda \supset \langle T_p Z, B \rangle$$

*with equations  $\lambda_1, \dots, \lambda_r$ . Suppose there is an open  $U_p \subset Z$  on which  $\pi_p$  is an isomorphism and a geometric point  $x \in (\Lambda \cap U_p)(k^s)$  in the generically free locus such that the matrix*

$$A(x)_{ij} = \lambda_i(x_j), \quad x_j \in V_{w_j}, \quad (4)$$

*has rank  $r$  and all  $r+1$  maximal minors nonzero. Then the Rosenlicht quotient  $Z/T$  is  $k$ -rational.*

*Proof.* The maximal minors first produce a unique point of  $\Lambda$  on a general  $T$ -orbit. The hypothesis that this section meets  $U_p$  then identifies it birationally with projective space.

For a geometric point  $q = \sum q_j + q_B$ , where  $q_j \in V_{w_j}$  and  $q_B \in \widehat{B}$ , put  $A(q)_{ij} = \lambda_i(q_j)$  and define

$$\kappa_j(q) = (-1)^j \det A(q)_{\widehat{j}}, \quad 0 \leq j \leq r. \quad (5)$$

The equations of  $\Lambda$  vanish on  $B$ , so  $tq \in \Lambda$  is equivalent to

$$A(q)(\chi^{w_0}(t), \dots, \chi^{w_r}(t))^t = 0.$$

The cofactor identity says that the kernel is generated by  $(\kappa_0(q), \dots, \kappa_r(q))$ . The basis hypothesis gives a unique  $t(q) \in T(k^s)$  through

$$\chi^{w_j - w_0}(t(q)) = \frac{\kappa_j(q)}{\kappa_0(q)}, \quad 1 \leq j \leq r. \quad (6)$$

These are Laurent monomials in the displayed ratios. The construction is unchanged if all weights are translated by one character, since only their differences occur. It is also independent of the ordering of the selected weight spaces: intrinsically,  $s(q) = t(q)q$  is the unique point of  $Tq \cap \Lambda$ . Because  $W$ ,  $\widehat{B}$ , and  $\Lambda$  descend, this uniqueness makes  $s$  Galois equivariant. Thus  $s$  is a  $k$ -rational, orbit-invariant retraction on the open where the action is free and the minors do not vanish. Let  $C$  be the closure of its image. Then  $C \dashrightarrow Z/T$  is birational and  $C \subset \Lambda \cap Z$ . Since  $x \in \Lambda$ , the vector  $(1, \dots, 1)$  lies in the kernel of  $A(x)$ ; uniqueness gives  $s(x) = x$ . Thus  $C$  meets  $U_p$ .

The image of  $\Lambda$  under  $\pi_p$  is a linear  $\mathbf{P}^{n-r} \subset \mathbf{P}^n$ , because  $\Lambda$  contains the centre of the linear projection. The irreducible variety  $C$  has dimension  $n - r$ , and  $\pi_p$  is injective on the nonempty open  $C \cap U_p$ . Consequently

$$\pi_p|_C : C \dashrightarrow \mathbf{P}^{n-r} \quad (7)$$

is birational. Hence  $Z/T$  is  $k$ -rational.

We record the inverse and its domain. Choose linear forms  $\rho_0, \dots, \rho_{n-r}$  whose restrictions give homogeneous coordinates on the image of  $\Lambda$  in  $\mathbf{P}(V/\hat{T}_p Z)$ . Let  $U \subset Z$  be the invariant open where the action is free, all  $\kappa_j(q)$  are nonzero,  $s(q) \in U_p$ , and  $\rho_0(s(q)) \neq 0$ . These conditions are invariant because  $s$  is orbit invariant and nonvanishing of the signed minors is invariant under the diagonal torus action. Then

$$\Phi : U/T \longrightarrow \mathbf{P}^{n-r}, \quad [q] \longmapsto [\rho_0(s(q)) : \dots : \rho_{n-r}(s(q))] \quad (8)$$

is the quotient parametrization. If  $R_\alpha(x)$  are homogeneous equations for  $Z$ , the graph of its inverse over  $\Phi(U/T)$  is the closure of the locus with  $x \in C \cap U_p$  satisfying

$$\begin{aligned} R_\alpha(x) &= 0 \quad \text{for every } \alpha, \\ \lambda_i(x) &= 0 \quad 1 \leq i \leq r, \\ y_j \rho_0(x) - y_0 \rho_j(x) &= 0 \quad 1 \leq j \leq n - r. \end{aligned} \quad (9)$$

Write its unique point over  $y$  as  $x(y)$  and set  $\Theta(y) = [x(y)]$ . The last line gives  $\Phi(\Theta(y)) = y$ . Conversely,  $s(q) \in \Lambda \cap U_p$  and has the same tangent-projection coordinates as  $x(\Phi([q]))$ ; injectivity of  $\pi_p|_{U_p}$  gives

$$\Theta(\Phi([q])) = [s(q)] = [q] \quad \text{in } U/T.$$

Thus both composites are the identity on the stated dense opens. In the Cox application, the  $R_\alpha$  in (9) are the twenty quadrics in [4, Section 3]. Equations (6) and (9) give the rational map and its inverse without an unrecorded finite cover.  $\square$

The integral-basis condition is essential. If the weight differences span a proper finite-index sublattice, the same equations generally define a finite cover of a  $T$ -orbit rather than a rational section.

The proof is constructive once equations for  $Z$ , a tangent point, and the linear forms  $\lambda_i$  are given. The signed minors in (5) compute the orbit correction, and (9) recovers the inverse parametrization by elimination. Thus a tangent-section witness produces explicit rational maps in either direction. No estimate for the complexity of that elimination is asserted.

### 3 The rank-three subtorus of the projective Cox model

Let  $T$  be the Néron–Severi torus, so  $X^*(T_k^-) = \text{Pic}(S_k^-)$ . The Galois-invariant cocharacter

$$\lambda_{-K} : \mathbf{G}_m \longrightarrow T, \quad D \longmapsto (-K_S, D), \quad (10)$$

acts by simultaneous scalar multiplication on the sixteen Cox generators. Write  $T_0 = T/\lambda_{-K}(\mathbf{G}_m)$ , a torus of rank five. Over  $\bar{k}$ , the projective Cox model

$$Z = \text{Proj Cox}(S) \subset \mathbf{P}^{15}$$

is a seven-dimensional variety with one apparent double point [4, Lemma 3.2].

Use the marking of [4, Sections 3–4]. For the type  $I_3$  Galois image, the sublattice

$$N_3 = \mathbf{Z}e_3 \oplus \mathbf{Z}e_4 \oplus \mathbf{Z}e_5 \subset X_*(T_{0,\bar{k}}) \quad (11)$$

is saturated and Galois stable. The two generators act on this basis by

$$\begin{bmatrix} -1 & 0 & 0 \\ -1 & 0 & 1 \\ -1 & 1 & 0 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & -1 \end{bmatrix}.$$

Let  $T_3 \subset T_0$  be the corresponding rank-three subtorus. Four blocks of Cox generators have the following projective  $T_3$ -weights:

| weight      | Cox generators           |
|-------------|--------------------------|
| $(0, 1, 1)$ | $L_{13}, L_{23}, L_{35}$ |
| $(1, 0, 1)$ | $L_{14}, L_{24}, L_{45}$ |
| $(1, 1, 0)$ | $E_1, E_2, E_5$          |
| $(1, 1, 1)$ | $L_{12}, L_{15}, L_{25}$ |

The four blocks form a Galois-stable set, but they are not individually Galois stable. On their displayed weights, the first generator acts by

$$(011 \ 111)(101 \ 110),$$

and the second by  $(110 \ 111)$ , fixing 011 and 101. These are affine actions on projective weights: a different diagonal lift translates all four weights by a common character. Every other Cox generator lies in the Galois-stable subspace

$$B = \mathbf{P}\langle E_3, E_4, L_{34}, Q \rangle. \quad (12)$$

The determinant of the three weight differences from  $(0, 1, 1)$  is one. Consequently the direct sum of the four selected blocks and the affine cone over  $B$  are complementary subspaces defined over the ground field, as required in Theorem 2.1. On the open where the four selected block components are nonzero, a projective stabilizer fixes their three ratios. The three weight differences form a basis, so that stabilizer is trivial. In particular, the  $T_3$ -action is generically free.

*Remark 3.1* (Saturation and the projective kernel). In the type- $I_1$  description, the three visible sign cocharacters in the last three coordinates are

$$(0, 1, 1), \quad (1, 0, 1), \quad (1, 1, 0).$$

They span the even-sum sublattice of  $\mathbf{Z}^3$ , of index two. Its saturation is  $\mathbf{Z}^3$ , obtained by adjoining

$$(1, 1, 1) = \frac{1}{2}((0, 1, 1) + (1, 0, 1) + (1, 1, 0)).$$

The map from the three parameter copies of  $\mathbf{G}_m$  to their image in the projective torus has kernel  $\mu_2$ . The degree two obtained from the unsaturated parameters counts this kernel; it is not the degree of the Rosenlicht quotient. Saturation gives the integral-basis condition in Theorem 2.1.

**Proposition 3.2** (A tangent section for the four Galois types). *For a smooth geometric quartic del Pezzo Cox model, there exist a general tangent point  $p$ , an orbit-test point  $x$ , and a codimension-three linear subspace  $\Lambda \supset \langle T_p Z, B \rangle$  such that  $x \in \Lambda \cap U_p$  and the maximal-minor condition of Theorem 2.1 holds at  $x$  for  $T_3$ . If  $Z$  is the projective Cox model of a smooth quartic del Pezzo surface  $S/k$  with  $S(k) \neq \emptyset$ , and the Galois image has type  $I_0, I_1, I_2$ , or  $I_3$ , such a triple  $(p, x, \Lambda)$  exists over the ground field.*

*Proof.* Use the standard blowup points

$$(1 : 0 : 0), (0 : 1 : 0), (0 : 0 : 1), (1 : 1 : 1), (1 : a : b), \quad \Delta = ab(a-1)(b-1)(a-b) \neq 0. \quad (13)$$

For a tangent point with auxiliary coordinates  $z = (z_1, z_2, z_3)$ , the space of tangent hyperplanes containing  $B$  has dimension four. Evaluation on the four selected blocks of Cox generators gives a 4-by-4 matrix. The following exact choices are used;  $e = (e_1, \dots, e_5)$  and  $z'$  specify the orbit-test point at which the evaluation matrix is tested.

| $i$ | $z$        | $e$                | $z'$        |
|-----|------------|--------------------|-------------|
| 1   | (1, 3, 7)  | (2, 3, 5, 7, 11)   | (2, 4, 9)   |
| 2   | (2, 5, 11) | (3, 4, 7, 13, 17)  | (1, 6, 10)  |
| 3   | (3, 8, 13) | (5, 7, 11, 17, 19) | (2, 9, 15)  |
| 4   | (4, 9, 17) | (2, 5, 11, 19, 23) | (3, 10, 18) |

Let  $D_i$  be the determinant of this evaluation matrix, and let  $M_i$  be the fixed 8-by-8 Jacobian minor at the same tangent point. After discarding nonzero rational factors, their numerators are recorded in Appendix A. Witness  $i$  is valid precisely on the open  $D_i M_i \neq 0$ . The appendix proves, by eight localized cases and one explicit Bézout identity, that

$$\bigcup_{i=1}^4 \{D_i M_i \neq 0\} = \{\Delta \neq 0\}. \quad (14)$$

Thus every smooth geometric Cox model has a witness for which the tangent point is smooth and the four-block evaluation map is an isomorphism at the orbit-test point. Invertibility is open in the pair of points. Since the projective Cox model is geometrically integral, this nonempty open meets the open where  $p$  is general for tangent projection and, for that  $p$ , the open  $U_p$  on which  $\pi_p$  is an isomorphism. We may also require that the  $T_3$ -action be free at  $x$ .

Let  $H_p$  be the four-dimensional space of tangent hyperplanes containing  $B$ . Evaluation identifies  $H_p$  with the four selected block values at  $x$ . Hyperplanes in  $H_p$  that vanish at  $x$  form a three-dimensional subspace; choose a basis  $\lambda_1, \lambda_2, \lambda_3$  and let  $\Lambda$  be their common zero locus. Then  $x \in \Lambda \cap U_p$ . Under the evaluation isomorphism the three rows span the hyperplane  $u_0 + u_1 + u_2 + u_3 = 0$ , so their four maximal minors are all nonzero. Thus the point satisfying the minor condition is the same point that lies in the tangent-projection isomorphism open.

It remains to descend a good triple. Fix a twist of one of the four types. Because  $T_3$  and  $B$  are Galois stable, the preceding conditions define a nonempty Galois-invariant open in the product of the tangent-point and orbit-test loci. Its projection to the tangent-point factor is dense constructible and therefore contains a nonempty open.

Corollary 3.5 of [4] supplies the universal torsor whose fibre over a fixed point of  $S(k)$  is trivial. Its  $k$ -points are Zariski dense by [4, Lemma 2.1], and the same holds in the projective Cox model. Hence the tangent-point open has a  $k$ -point. At that point, density again supplies a  $k$ -rational orbit-test point in the corresponding nonempty open. The kernel of evaluation at that point is a three-dimensional  $k$ -space, so a  $k$ -basis gives  $\Lambda$ . No individual split-coordinate witness is asserted to descend.

For type  $I_3$ , stability of  $N_3$ ,  $B$ , and the set of four selected Cox blocks is checked on the two displayed generators. Up to conjugacy in  $W(D_5)$ , the groups of types  $I_0, I_1, I_2$  are subgroups of the group of type  $I_3$  [4, Lemma 4.2]. Restriction and a change of marking preserve the same data. This exhausts the four Galois images allowed by the hereditary  $H^1$ -condition; it is not a classification of arbitrary subgroups of  $W(D_5)$ .  $\square$

## 4 Proofs of the rationality statements

*Proof of Theorem 1.2.* We first treat the minimal case by taking a quotient of the projective Cox model. An equivariant generic trivialization then identifies that quotient with the product of  $S$  and a two-dimensional torus. The nonminimal case reduces to the classical rationality theorem in degree at least five.

Suppose first that  $S$  is  $k$ -minimal. Since a stably permutation module has vanishing  $H^1$  on every open subgroup, Proposition 4.1 and Lemma 4.2 of [4] place the Galois image in one of the four types used in Proposition 3.2. Lemma 3.2 and Theorem 2.4 of [4], together with Theorem 2.1, give

$$Z/T_3 \sim_k \mathbf{P}^4. \quad (15)$$

By [4, Corollary 3.5], let  $\mathcal{T} \rightarrow S$  be a universal torsor under the Néron–Severi torus  $T$ , chosen to have a  $k$ -point above a fixed point of  $S(k)$ . The stable-permutation hypothesis and Hilbert’s Theorem 90 give

$$H^1(F, T) = 0 \quad \text{for every field extension } F/k \quad (16)$$

by [4, Remark 2.2]. Thus the generic fibre over  $k(S)$  has a point. Translation by the resulting rational section  $\sigma$  defines the  $T$ -equivariant birational map

$$S \times T \dashrightarrow \mathcal{T}, \quad (s, t) \mapsto \sigma(s)t. \quad (17)$$

The split scalar subtorus in (10) acts as common multiplication on every Cox coordinate. The universal torsor is a dense open in the affine Cox cone, and its scalar quotient is a dense open in  $Z$ . Quotienting (17) by that scalar therefore gives a  $T_0$ -equivariant birational map  $S \times T_0 \dashrightarrow Z$ . Since  $T_3 \subset T_0$  is a  $k$ -subtorus, Rosenlicht quotients give

$$Z/T_3 \sim_k S \times (T_0/T_3). \quad (18)$$

The quotient  $T_0/T_3$  is a two-dimensional  $k$ -torus, hence is  $k$ -rational by [6]. Equations (15) and (18) prove that  $S \times \mathbf{A}^2$  is  $k$ -rational.

If  $S$  is not  $k$ -minimal, successively contracting  $k$ -defined unions of pairwise disjoint exceptional curves gives a del Pezzo surface  $S'$  of degree at least five. The image of a  $k$ -point of  $S$  is a  $k$ -point of  $S'$ , so  $S'$ , and therefore  $S$ , is  $k$ -rational by [5, Theorem 2.1].

Finally, if  $S$  is stably  $k$ -rational, a nonempty open of a projective space is birational to a nonempty open of  $S \times \mathbf{P}^m$ . Since  $k$  is infinite, this supplies a  $k$ -point on  $S$ . Manin’s stable birational Picard-module obstruction implies that  $\text{Pic}(S_{\bar{k}})$  is stably permutation. In the form used here, this consequence of [2, Theorem 2.2] is recalled immediately before [4, Proposition 4.1] and included in [4, Corollary 4.3]. This proves the last assertion.  $\square$

*Proof of Corollary 1.3.* The argument takes place entirely over the function field of the projection base. This avoids any specialization or fibrewise rationality claim.

For  $X = X_{j,r}$ , Tschinkel and Zhang project from a plane containing a distinguished line. The base is  $\mathbf{P}_{\mathbf{Q}}^{r+1}$ , with function field  $K = \mathbf{Q}(t_1, \dots, t_{r+1})$ , and the generic cubic-surface fibre is  $K$ -birational, after contracting the distinguished line, to a smooth quartic del Pezzo surface  $S/K$  with a  $K$ -point. Their calculation identifies the Galois image as type  $I_1$  for  $j = 1$  and type  $I_3$  for  $j = 3$  [4, Propositions 5.1–5.2]. Lemma 4.2 of [4] then shows that  $\text{Pic}(S_{\bar{K}})$  is stably permutation. Theorem 1.2 gives

$$K(S)(u, v) \simeq K(z_1, z_2, z_3, z_4).$$

Since  $\mathbf{Q}(X) = K(S)$  as extensions of  $K$ ,

$$\mathbf{Q}(X)(u, v) \simeq \mathbf{Q}(t_1, \dots, t_{r+1}, z_1, z_2, z_3, z_4)$$

is purely transcendental. This is rationality of  $X \times \mathbf{A}^2$ , equivalently of  $X \times \mathbf{P}^2$ . This is a function-field argument; no specialization assertion is used.  $\square$

*Proof of Theorem 1.1.* Version 0.15.0 of the one-stabilization manuscript proves that  $X \times \mathbf{P}^1$  is irrational for every smooth complex cubic threefold  $X$  [3, Theorem 1.1]. It applies to  $(X_j)_{\mathbf{C}}$ . Therefore neither  $X_j$  nor  $X_j \times \mathbf{P}^1$  can be rational over  $\mathbf{Q}$ , because rationality survives extension of the ground field. Corollary 1.3 gives rationality at  $m = 2$  over  $\mathbf{Q}$ , and hence also over  $\mathbf{C}$ . This proves both equalities of stabilization levels.  $\square$

*Proof of Corollary 1.4.* Theorem 1.1 shows that  $Y_j = X_j \times \mathbf{P}^1$  is nonrational, even after base change to  $\mathbf{C}$ . On the other hand,

$$\mathbf{Q}(Y_j \times \mathbf{A}^1) = \mathbf{Q}(X_j)(u, t)$$

is purely transcendental by Corollary 1.3. Equivalently,  $\mathbf{P}^1 \times \mathbf{A}^1$  is birational to  $\mathbf{A}^2$ . Hence  $Y_j \times \mathbf{A}^1$  is rational.  $\square$

*Proof of Corollary 1.5.* Theorem 1.1 gives the value two on each isomorphism class represented by the displayed examples. Engel, de Gaay Fortman, and Schreieder prove that a very general smooth complex cubic threefold does not admit a decomposition of the diagonal and hence is not stably rational [1, Corollary 1.4]. For such a threefold no finite  $m$  makes  $X \times \mathbf{P}^m$  rational, so  $\ell_{\mathbf{C}}(X) = \infty$ .  $\square$

## A Exact tangent-section certificate

This appendix records the exact cover used in Proposition 3.2. It exposes the polynomials and the two surviving localized branches used in the geometric nonemptiness claim. The public generator proves the emptiness of six branches by exact Gröbner computations. For each branch, the public certificate also retains a machine-readable identity expressing a nonzero constant in the corresponding localized ideal. The displayed evaluations and Bézout identity exclude the two surviving branches.

After discarding nonzero rational factors, the four evaluation determinants are

$$\begin{aligned} D_1 &= (8a - 27)^2 (3968a - 1349b - 2619), \\ D_2 &= 568453977a^3 - 373279764a^2b - 1658788404a^2 \\ &\quad + 59068867ab^2 + 976763788ab + 925668016a \\ &\quad - 128537024b^2 - 359163728b - 10185728, \\ D_3 &= 1643918400a^3 - 1105797420a^2b - 7261914240a^2 \\ &\quad + 154718524ab^2 + 3804690316ab + 9944961600a \\ &\quad - 282623593b^2 - 3443769343b - 3454184244, \\ D_4 &= 35068545a^3 + 2864538a^2b - 236376549a^2 \\ &\quad - 5350827ab^2 + 24417420ab + 469658574a \\ &\quad + 14536855b^2 - 90750394b - 214068162. \end{aligned}$$

and the corresponding smoothness minors are

$$\begin{aligned} M_1 &= -4(a-1)(b-7)(3a-b-2)(2ab+7a-9b), \\ M_2 &= -9(a-1)(2b-11)(3a-b-2)(4ab+11a-15b), \\ M_3 &= -25(a-1)(3b-13)(2a-b-1)(3ab+13a-16b), \\ M_4 &= -(a-1)(4b-17)(13a-5b-8)(32ab+85a-117b). \end{aligned}$$



To prove (14), suppose first that  $D_i M_i = 0$  for  $i = 1, 2, 3$ . Choosing one factor from each product gives eight cases. A lexicographic Gröbner calculation in  $\mathbf{Q}[a, b, h]/(h\Delta - 1)$  gives

| chosen equations                                     | localized zero locus                  |
|------------------------------------------------------|---------------------------------------|
| $D_1 D_2 D_3, D_1 D_2 M_3, D_1 M_2 D_3, D_1 M_2 M_3$ | empty                                 |
| $M_1 D_2 D_3, M_1 D_2 M_3$                           | empty                                 |
| $M_1 M_2 D_3$                                        | $3a - b - 2 = Q_0(b) = 0$             |
| $M_1 M_2 M_3$                                        | $3a - b - 2 = (3b - 26)(3b - 13) = 0$ |

where

$$Q_0(b) = 31223016b^2 - 435944529b + 1306078948.$$

On the line  $3a - b - 2 = 0$ , the primitive numerator of  $D_4 M_4$  is

$$-(b-1)^4(4b-17)(16b-85)(83246b^2 - 872181b + 2185995),$$

$$Q_4(b) = 83246b^2 - 872181b + 2185995.$$

The evaluations

$$Q_0(17/4) = \frac{69121705}{4}, \quad Q_0(85/16) = -\frac{4117757269}{32},$$

$$Q_4(26/3) = \frac{7918133}{9}, \quad Q_4(13/3) = -\frac{272530}{9}$$

exclude the four linear roots. Finally, the two quadratics are coprime by the identity

$$(2265746679974131615 - 377042650728395274b)Q_0(b) \\ + (141417109727495582904b - 1342668830289072756147)Q_4(b) \\ = 24176690547344887359179755.$$

Thus  $D_4 M_4 \neq 0$  in both remaining cases. This proves that the four opens  $D_i M_i \neq 0$  cover  $\Delta \neq 0$ .

## B Verification and scope

We compute the cocharacter saturation, type- $I_3$  actions, Cox weights, determinant-one weight differences, and tangent-incidence cover using exact integer and rational arithmetic. The public generator starts from the two type- $I_3$  character matrices and a transcription of the twenty Cox quadrics in [4, Section 3]. It reconstructs the saturated rank-three lattice, the residual rank-two quotient, the tangent Jacobian, all four pairs  $(D_i, M_i)$ , the eight localized branches, and the Bézout identity above with SymPy 1.14.0. The certificate retains exact Nullstellensatz identities for the six empty branches; both public programs verify those identities directly. The generator also derives the evaluation-kernel slice through the orbit-test point and renders every computation-derived matrix, block, witness, determinant, minor, and Bézout coefficient printed in the manuscript into a checked TeX artifact. A second checker independently verifies the resulting cover, kernel slice, and quotient minors. We check the transcription of the source equations, the quotient criterion, equivariant generic trivialization, incidence descent, and function-field argument in the text; the programs do not address them.

No Lean development formalizes Theorem 1.1, Theorem 1.2, Theorem 2.1, Proposition 3.2, or the stated corollaries. The formal-coverage annotations therefore say **absent**.

We import Tschinkel–Zhang’s universal-torsor geometry and their classification of the four hereditary- $H^1$  Galois images. We construct the rational rank-three quotient and its descended tangent section. This construction lowers the number of stabilizing variables to two; no classification beyond those four Galois types is needed here.

## AI assistance disclosure

This project was developed with extensive assistance from OpenAI Codex and Anthropic Claude. Under the author’s direction, the systems assisted with proof exploration, literature research, exact computation, verification, and manuscript drafting and revision. The author checked the resulting arguments, computations, code, and cited sources, reviewed and edited all AI-assisted material, and assumes responsibility for all content.

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